MIL-S-20322B
19 April 1973
SUPERSEDING
MIL-S-20322A
17 April 1956

MILITARY SPECIFICATION

STRONTIUM NITRATE, ANHYDROUS

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers anhydrous strontium nitrate for use in pyrotechnic compositions.

1.2 Classification. Strontium nitrate, anhydrous shall be of the following grades, as specified (see 6.1 and 6.2):

Grade A
Grade B

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
- L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
- RR-S-366 - Sieve, Test
- PPP-D-723 - Drums, Fiber

MILITARY
- MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
- MIL-D-6054 - Drums, Metal - Shipping and Storage.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

D 1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Chemical and physical requirements. The strontium nitrate shall conform to the chemical and physical requirements specified in table I.

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TABLE I. Chemical and Physical Properties

Property	Requirement		Test paragraph
	Grade A	Grade B	
Strontium nitrate (including any barium nitrate), %, min.	99.5	99.5	4.4.1
Moisture, %, max.	0.05	0.05	4.4.2
Hygroscopicity, %, max.	0.05	0.05	4.4.3
Acidity (pH)	6.0-7.0	6.0-7.0	4.4.4
Water-insoluble matter, %, max.	0.02	0.02	4.4.5
Grit, %, max.	0.02	0.02	4.4.6
Chloride (as Cl), %, max.	0.003	0.003	4.4.7
Ammonium compounds (as NH_3), %, max.	0.01	0.01	4.4.8
Sulfur (total), %, max.	0.01	0.01	4.4.9
Barium, %, max.	0.05	1.5	4.4.10
Calcium, %, max.	0.003	0.003	4.4.10
Magnesium, %, max.	0.003	0.003	4.4.10
Sodium, %, max.	0.003	0.003	4.4.10
Iron, %, max.	0.003	0.003	4.4.10
Bismuth, %, max.	0.003	0.003	4.4.10
Lead, %, max.	0.003	0.003	4.4.10
Copper, %, max.	0.003	0.003	4.4.10

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3.2 Granulation

3.2.1 Grade A. The granulation requirement for Grade A strontium nitrate shall be 95 percent through a number (No.) 100 U. S. standard sieve, min., when tested in accordance with 4.4.11. The sieve used shall conform to RR-S-366.

3.2.2 Grade B. The granulation requirements for Grade B strontium nitrate shall be 99.9 percent through a No. 50 U. S. standard sieve, min., and 70.0 percent retained on a No. 140 sieve, min. The sieves used shall conform to RR-S-366.

3.3 Workmanship. The material shall be free from grit, dirt, foreign matter and other injurious materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The test procedures specified hereinafter shall be used in determining compliance of the material with the corresponding requirements specified in section 3. Failure of a specimen to comply with any requirement of the acceptance test shall result in the rejection of the lot represented.

4.2 Lotting. A lot shall consist of strontium nitrate produced consecutively on one shift, by one manufacturer, at the same plant, using the same equipment, under essentially the same manufacturing conditions, with no change in raw materials, provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.3 Sampling. Containers shall be randomly selected from each lot in accordance with MIL-STD-105, inspection level S-4, with a minimum sample size of five containers per lot. The contents of each container shall be thoroughly mixed and approximately 8 ounces of the material removed from each of the selected containers. Each primary sample shall be placed in an air-tight container, labeled so that the container from which it was taken can be easily identified. A composite sample of approximately 16 ounces shall be made from equal

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portions of the primary samples and mixed thoroughly. The composite sample shall be placed in an air-tight container, labeled to show the name of the material, manufacturer, plant, contract or purchase order, batch number, and lot number. All acceptance tests shall be made on specimens selected from the composite sample representative of the lot. However, if it becomes apparent during sampling that the lot is not uniform throughout, the inspector may require that any of the primary samples be tested for compliance with the requirements of this specification. The primary samples shall be held for future examination should the composite sample fail to meet the specification requirements.

4.4 Tests. Distilled water in accordance with ASTM D 1193 Type III, and reagent grade chemicals shall be used throughout the tests. Blank determinations shall be run and corrections shall be applied when necessary.

4.4.1 Strontium nitrate. Place 10.0 ±0.5 gm. of tungstic oxide into a platinum crucible and ignite at 700° ±50°C (1292° ±90°F) in a furnace for 1 hour. Cool in a desiccator for 60 ±5 minutes and weigh. Place the crucible back into the furnace for an additional half hour, cool in a desiccator for 60 ±5 minutes, and weigh again. Repeat the procedure until constant weight is obtained. Allow the furnace to cool to approximately room temperature and adjust the controls to 690°C (1274°F). Mix a 2 gm. portion of the sample (accurately weighed to 0.1 mg.) with the ignited tungstic oxide and cover the crucible with a tight fitting cover. Place the crucible in the furnace at approximately room temperature and heat to 690°C (1274°F). Heat the crucible for 2 hours after the operating temperature has been reached, cool in a desiccator for 60 ±5 minutes, and weigh. Reheat for an additional half hour, cool in a desiccator for 60 ±5 minutes, and weigh again. Repeat the procedure until constant weight is obtained. Calculate the percent strontium nitrate as follows:

$$\text{Percent strontium nitrate} = \frac{195.9A}{W}$$

where, A = loss in weight, gm.
W = weight of sample, gm.

4.4.2 Moisture. Transfer 3 gm. of the sample to a tared weighing dish (approximately 6 cm. in diameter and 2.5 cm. deep) and weigh. Dry at 130° ±3°C (265° ±5°F) for 3 hours, cool in a desiccator, and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\ A}{W}$$

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where, A = loss in weight, gm.
W = weight of sample, gm.

Save the dried samples for the hygroscopicity determination (see 4.4.3).

4.4.3 Hygroscopicity. Place the dried samples from the moisture determination into a desiccator, 10 liter capacity, containing 1 liter of a sulfuric acid solution (33.1 ±0.2% sulfuric acid by weight) for producing 70% relative humidity at 25°C (77°F). Allow the dishes to remain in the desiccator for 48 hours and weigh. Place the dishes back into the desiccator for 24 hours and weigh again. Repeat for 24 hour intervals until constant weight is obtained. Usually the weight will not change significantly after the initial 48 hour period. Calculate the percent hygroscopicity as follows:

$$\text{Percent hygroscopicity} = \frac{100\ A}{W}$$

where, A = gain in weight, gm.
W = weight of sample, gm.

4.4.4 Acidity (pH). Dissolve 5 gm. of the sample in 100 ml. of distilled water. Determine the pH of the solution at 25°C (77°F).

4.4.5 Water-insoluble matter. Weigh 50 gm. of the sample into a 400 ml. beaker, add about 300 ml. of water, and stir to dissolve. Heat to boiling and boil for 5 minutes with occasional stirring. Filter through a tared sintered glass crucible and wash with hot water. Dry the crucible at 105° ±3°C (221° ±5°F) for 1 hour, cool in a desiccator, and weigh. Calculate the percent water-insoluble matter as follows:

$$\text{Percent water-insoluble matter} = \frac{100\ A}{W}$$

where, A = weight of residue, gm.
W = weight of sample, gm.

4.4.6 Grit. Weigh 50 gm. of the sample into a 400 ml. beaker, add about 300 ml. of water, and stir to dissolve. Heat to boiling and boil for 5 minutes with occasional stirring. Filter through a Whatman No. 40 filter paper and transfer and wash with hot water. Save the filtrate for the determination of chloride (see 4.4.7). Wash the residue from the filter paper into a 100 ml beaker with a stream of water from a wash bottle. Add 10 ml. of nitric acid to the 100 ml beaker and evaporate to dryness on a steam bath. Add about 10 ml. of water and break up

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any residue with a policeman. Transfer to a No. 170 U. S. standard sieve with water. If any solid material is retained by the sieve, wash well with water, alcohol, and finally ethyl ether. Transfer the material to a tared weighing dish and weigh. Calculate the percent grit as follows:

$$\text{Percent grit} = \frac{100\ A}{W}$$

where, A = weight of residue, gm.
W = weight of sample, gm.

4.4.7 Chloride. Add 5 ml. of nitric acid to the filtrate from the grit determination (see 4.4.6). Heat to boiling, add 5 ml. of 5% silver nitrate solution, and boil for 5 minutes. Allow to stand in a dark place overnight at room temperature. Filter through a tared sintered glass crucible of fine porosity and wash with water. Dry at 150° ±5°C (302° ±9°F) for 1 hour, cool in a desiccator, and weigh. Calculate the percent chloride as follows:

$$\text{Percent chloride (as Cl)} = \frac{24.7A}{W}$$

where, A = weight of percipitate, gm.
W = weight of sample, gm.

4.4.8 Ammonium compounds. Weigh 10 gm. of the sample into a 500 ml. Kjeldahl flask and dissolve in 100 ml. of water. Add 10 ml. of 20% sodium hydroxide solution, immediately connect to the Kjeldahl distillation apparatus, the outlet tube of which dips below 50 ml. of 3% boric acid solution contained in a 250 ml. Erlenmeyer flask. Distill about 75 ml., wash down the outlet tube, add about 3 drops of methyl purple indicator, and titrate with standard 0.05 N sulfuric acid solution until the color changes from green to light purple. Carry a blank determination through the entire procedure. Calculate the percent ammonium compounds as follows:

$$\text{Percent ammonium compounds (as } NH_3) = \frac{1.70\ (A-B)\ N}{W}$$

where, A = sulfuric acid for titration of sample, ml.
B = sulfuric acid for titration of blank, ml.
W = weight of sample, gm.
N = normality of sulfuric acid

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4.4.9 Sulfur (total). Weigh 20 gm. of the sample into a 400 ml. beaker, add about 200 ml. of water, and stir to dissolve. Add 2 ml. of hydrochloric acid and 10 ml. of bromine water. Heat to boiling and boil 10 minutes. Filter through a Whatman No. 40 filter paper and wash with water. Add 10 ml. of 10% barium chloride solution, heat to boiling, and boil 5 minutes. Allow to stand overnight. Filter through a Whatman No. 42 filter paper, transfer, and wash with water. Place the filter paper and precipitate into a tared platinum crucible. Char and burn off the filter paper at dull red heat and then ignite at red heat for 30 minutes. Cool in a desiccator and weigh. Calculate the percent sulfur as follows:

$$\text{Percent sulfur (total)} = \frac{13.7\ A}{W}$$

where, A = weight of precipitate, gm.
W = weight of sample, gm.

4.4.10 Barium, calcium, magnesium, sodium, iron, bismuth, lead, and copper. Determine the calcium, barium, magnesium, sodium, iron, bismuth, lead and copper contents by spectrographic or other instrumental methods approved by the contracting officer.

4.4.11 Granulation. Dry the material to be sieved by spreading in a thin layer on a large watch glass and heating in an oven at 57° ±3°C (135° ±5°F) for 1 hour. Approximately 100 gm. of dried Grade A material shall be transferred to a No. 100 U. S. standard sieve. Approximately 100 gm. of dried Grade B material shall be transferred to a No. 50 U. S. standard sieve nested over a No. 140 sieve. The sieve or sieve assembly shall be provided with a cover and a bottom pan. The sieves shall conform to RR-S-366. Shake for 10 minutes by hand, or for 5 minutes by means of a mechanical shaker geared to produce 300 ±15 gyrations and 150 ±10 taps of the striker per minute. Weigh the retained portions on each sieve and calculate to percent passed or retained, as required.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Strontium nitrate shall be packed in 11 gallon (130 lb.) steel drums conforming to MIL-D-6054. Steel drums shall be with full open

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head provided with a twist lock closure or bolted ring closure. The lid shall have a tubular rubber gasket. The drums shall be provided with a bag liner fabricated from nominal 6 ml. thick polyethylene. Polyethylene bags shall be made from plastic film conforming to L-P-378, type I, finish 1. All bag seams and closure shall be heat sealed. Three units of desiccant conforming to MIL-D-3464 shall then be put on top of the sealed polyethylene bag (to absorb any moisture which may be entrapped). A printed card identified with marking in accordance with MIL-STD-129 shall be inserted just before the drum is closed.

5.1.2 Level B. Strontium nitrate shall be packed in fiber drums conforming to PPP-D-723, Type I, grade A, class 2; Type II, grade A; and Type III, grade A, class 2, as applicable. Polyethylene bag as specified in 5.1.1 shall be provided. Fiber drums shall be with full open head provided with a locking ring closure. Net weight of the material of each drum shall be 150 pounds.

5.1.3 Level C. Unless otherwise specified (see 6.2), 150 pounds of strontium nitrate shall be packed in a container constructed to assure that the strontium nitrate will remain in prime condition and adequate to assure safe delivery to destination without damage. The container shall comply with Uniform Freight Classification, National Motor Freight Classification, and Department of Transportation Regulations, as applicable. The container shall be sufficiently rigid to withstand transportation and to permit storing in tiers at least 10 feet high without damage to containers or contents. The gross weight and size limitations of the containers shall not exceed the tariff or classification limitations specified therein, and the containers shall be closed or strapped as specified therein.

5.2 Marking. The metal drums shall be marked in accordance with Department of Transportation Regulations, and with MIL-STD-129.

5.2.1 Packs prepared for level C shall be conspicuously marked MIN MIL PK. When specified, containers for overseas shipment shall be marked with the Ordnance Corps symbol, the SNL group of the item and the method of preservation.

6. NOTES

6.1 Intended use. Strontium nitrate Grade A and Grade B are for use as follows:

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Grade A - For use in pyrotechnic compositions, such as flares.
Grade B - For use in tracer compositions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Grade required (see 1.2).
c. Level of packing required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or has been subjected to some physical mixing operation intended to make the final product substantially uniform.

Custodians:
Army - MU
Navy - AS

Review activities:
Army - MU, MD

User activities:
Army - WC
Navy - AS
DSA - GS

Preparing activity:
Army - MU

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